John Aziz

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SUMMARY

Senior AI Software Engineer who has shipped production software since 2021, building AI systems, Python backends and cloud-native applications. Maintainer of CacheLib and former Microsoft MVP.

PROFESSIONAL EXPERIENCE

Lyah Dec 2025 - Present

Senior AI Software Engineer (Python, Vue.js)

- *Designed and built the modular, node-based LangGraph framework behind Lyah's course generation, tutoring and copilot agents, replacing legacy n8n workflows, with automatic fallback across OpenAI, Anthropic and Google, Celery orchestration and Opik observability.*
- *Replaced the single-tenant model with multi-tenant organizations in production: scoped roles, a required authorization policy on every route and socket event, session-backed auth and an append-only audit log.*
- *Built a cost ledger that attributes vendor calls (LLMs, voice, storage, payment fees) to an organization, course and person, and reports margin in EUR and USD.*
- *Added retries that resume a failed session workflow from the stage that failed.*
- *Built course and lecture generation agents, the Vue 3 authoring tools with AI proposal review, a RAG tutor over WebSocket chat, and instructor analytics reports.*
- *Built an evaluation and prompt-optimization pipeline for the slide-transition agent (LiteLLM, Opik MetaPrompt), comparing OpenAI, Anthropic and Google models across 71 experiments. On a hand-labeled production set, the production model reached 93% accuracy with no false slide changes, and stayed 92% consistent on recent traces.*
- *Built Stripe course billing and usage limits by session and course time, and integrated Moodle and Canvas through LTI, including grade passback and embedded launches.*
- *Extended the Pipecat voice sessions with reconnection and concurrent-session handling, step-based progress tracking and per-service usage reporting.*

Orange Business Nov 2024 - Dec 2025

AI Software Engineer, Egypt AI Factory (Oct 2025 - Dec 2025)
Software and Data Engineer, International Data and Automation (Nov 2024 - Oct 2025)

- *Managed two senior engineers in India (Jun 2025 - Oct 2025), until a restructure moved me to the AI Software Engineer role.*
- *Built RAG and document ingestion pipelines for contracts and SharePoint using pgvector, OpenAI and scheduled batch jobs.*
- *Responsible for 2 production applications and led development on a third, including one as sole developer, and managed their migration to a new cloud environment.*
- *Published an internal Python package used across applications, and deployed services with Kubernetes, Helm and GitLab CI/CD.*
- *Delivered backend and frontend features with FastAPI and React.js, working from requirements gathered with business stakeholders.*
- *Integrated OIDC authentication and added Redis caching to improve API performance.*
- *Provided code reviews and technical guidance across the team, and supported production deployments and live issues.*

Manning Publications Jun 2024 - Present

Freelance Technical Proofreader

- *Validated code for* Generative AI in Action *and* LLM Customization and Fine-Tuning*, including fixes merged into a book's companion repository.*

Fabrica Dev Aug 2024 - Nov 2024

Software Engineer (Python, Vue.js)

- *Built an Egyptian Tax Authority integration to generate and submit digital invoices.*
- *Built backend services and features with Python, MySQL, MongoDB and Vue.js.*

GEMINDZ Mar 2024 - Aug 2024

Software Developer (Python, Vue.js)

- *Built a Python package to manage multi-state hardware devices and automate firmware flashing, and automated on-device testing with Pytest.*
- *Built and deployed a full-stack web application with FastAPI, Vue.js and Azure.*
- *Built OS-independent monitoring utilities and integrated them with distributed software.*

Udacity and iSchool Jun 2022 - Apr 2024
Technical Trainer and Mentor
- *Taught full-stack Python, computer science and programming, and reviewed student projects.*

Arram Group Jul 2021 - Jan 2022
Software Engineer (Python, Flask)
- *Sole developer of a Flask CRM used by all 14 employees, from database to deployment.*
- *Automated data processing with Pandas, and maintained and improved the company's 3 websites.*

Open Source Jan 2021 - Present
Maintainer and Contributor
- *CacheLib (Pallets Eco, maintainer since Apr 2026): restarted releases for the Python caching library behind Flask-Caching and Flask-Session, with 7.4M monthly downloads, shipping 9 releases in under 4 months, adding a Valkey backend and a JSON serializer, and fixing Redis and memcached counter bugs.*
- *markdown-checker (author): link validator and GitHub Action used in Microsoft's Generative AI for Beginners (123k stars), Azure Search OpenAI Demo (7.8k stars) and BeeWare's docs tooling, with 45k downloads in its peak month (Jun 2026).*
- *Flask-Caching (contributor, 4.5M monthly downloads): moved it onto the latest CacheLib, made FIPS-compatible hashing the default with a configurable hash_method, added memcached pool options, and set up CI against real backends.*
- *Developed the landing website for Azure SQL, attracting over 5k monthly views.*
- *Azure Developer CLI: implemented 2 features and fixed 1 bug in Go.*

EDUCATION

Modern Academy for Engineering and Technology Sep 2018 - Jul 2023
B.Sc. in Computer Engineering and Information Technology — CGPA: 3.44/4.0
- *Ranked 4th in my class in a cohort of over 150 students.*

PROFESSIONAL CERTIFICATIONS

Google Cloud Certified: Professional Cloud Architect — Feb 2024 - Feb 2028
Microsoft Certified: Azure AI Engineer Associate — Mar 2023 - Mar 2027
AWS Certified Cloud Practitioner — Feb 2023 - Dec 2028
Google Cloud Certified: Associate Cloud Engineer — May 2022 - May 2028

PUBLICATIONS

Microsoft Learn
Introduction to large language models — Dec 4, 2023

Microsoft Tech Community
Bring your own Data to Azure OpenAI: Step-by-Step Guide — Sep 7, 2023
Deploying a Large Language Model (GPT-2) on Azure Using Power Automate: Step-by-Step Guide — Jun 21, 2023

VOLUNTEERING EXPERIENCE

- *Microsoft AI MVP (Jan 2024), Top MENA GDSC Lead 2020-2021, and Microsoft Learn Student Ambassadors Technical Lead (Jan 2021 - Aug 2024).*
- *Wrote 12 Microsoft Tech Community articles and 3 Microsoft Learn modules (one since retired) with 150k+ combined views, and was featured in Microsoft's Contributor Stories.*

Leader *– Azure Egypt User Group* — Jan 2024 - Oct 2024
Organizer *– Google Developer Groups Cloud Egypt* — Jan 2021 - Jul 2024
Mentor *– Imagine Cup* — Dec 2022 - Feb 2023
Judge *– Imagine Cup Junior Competition* — May 2023
Technical Trainer *– BambooGeeks, Google IT certificate learners* — Feb 2022 - May 2022
Community Leader *– Google Developer Student Clubs* — Aug 2020 - Jul 2021
Vice President *– Enactus Modern Academy* — Aug 2019 - Aug 2020

LANGUAGES

Arabic (Native), English (Fluent), French (Elementary)